



Hamed Hosseini

AI & ROBOTICS RESEARCHER · LECTURER

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Researcher in robotic perception and manipulation, with work spanning grasp inference, geometry-aware scene abstraction, embodied AI, and computer vision. My research connects deep learning with physical systems, simulation, and real-robot evaluation.

RESEARCH AREAS

Robotic grasping and manipulation · Computer vision and RGB-D perception · Scene understanding · Embodied AI · Human–robot interaction · Deep learning for intelligent robotics

EDUCATION

Ph.D. in Artificial Intelligence and Robotics

2017–2025 · University of Tehran, Tehran, Iran · GPA 17.25/20

Thesis: **Intelligent Robotic Grasp Using Deep Neural Networks**. Research on object decomposition into 3D primitive shapes for grasp detection, with CoppeliaSim simulation and evaluation on a real parallel delta robot with a custom gripper. Supervisors: Dr. M. T. Masouleh and Dr. A. Kalhor.

M.Sc. in Electrical Engineering, Control

2015–2017 · Amirkabir University of Technology, Tehran, Iran · GPA 17.30/20

Thesis: Employing Biological Signals in the Rehabilitation Procedure.

B.Sc. in Electrical Engineering

2011–2015 · Amirkabir University of Technology, Tehran, Iran · GPA 16.12/20

Thesis: Artifact Detection at the EEG Signals.

ACADEMIC EXPERIENCE

Lecturer, Artificial Intelligence in Robotics and Mechatronics

University of Tehran · August 2023–present

Teaching artificial intelligence for robotics and mechatronics.

Research Assistant, Human–Robot Interaction Laboratory

University of Tehran · August 2017–present

Research in machine learning, robotics, and control, including deep-learning-based robotic grasping and scene rearrangement.

Research Assistant, Parsian Robotic Lab

Amirkabir University of Technology · December 2015–August 2017

Small-size robot soccer, defensive strategies, team coordination, and control.

SELECTED PUBLICATIONS

Hosseini, S., Koosheshi, M. H., Masouleh, M. T., & Kalhor, A. (2024). Multi-Modal Robust Geometry Primitive Shape Scene Abstraction for Grasp Detection. *IEEE Access*, 12, 130117–130134. DOI: 10.1109/ACCESS.2024.3458800.

Koosheshi, M. H., Hosseini, S., Masouleh, M. T., Kalhor, A., & Hairi Yazdi, M. R. (2023). AGILE: Approach-Based Grasp Inference Learned from Element Decomposition. *11th RSI International Conference on Robotics and Mechatronics*. DOI: 10.1109/ICRoM60803.2023.10412429.

Koosheshi, A. K. M. H., Hosseini, S., & Masouleh, M. T. (2025). Learning Post-approach Robotic Grasping using Convolutional Neural Networks. *13th RSI International Conference on Robotics and Mechatronics (ICRoM)*.

Nasr Esfahani, A., Hosseini, S., Masouleh, M. T., & Kalhor, A. (2026). Embodied AI for Kitchen Scene Rearrangement: A Deep Spatio-Temporal Approach. *Intelligent Service Robotics*, 19(4), 71.

Barghi, A., Hosseini, S., Ghasemi, S., Masouleh, M. T., & Kalhor, A. (2026). Dynamic Buffers: Cost-Efficient Planning for Tabletop Rearrangement with Stacking. *IEEE Transactions on Automation Science and Engineering*.

Ghasemi, S., Hosseini, S., Koosheshi, M. H., Masouleh, M. T., & Kalhor, A. (2024). Scene Understanding in Pick-and-Place Tasks: Analyzing Transformations Between Initial and Final Scenes. *32nd International Conference on Electrical Engineering*.

Nasr Esfahani, A., Hosseini, S., Masouleh, M. T., Kalhor, A., & Sajedi, H. (2024). AI-Driven Relocation Tracking in Dynamic Kitchen Environments. *14th International Conference on Computer and Knowledge Engineering*.

Hosseini, S., Masouleh, M. T., & Kalhor, A. (2020). Improving the Successful Robotic Grasp Detection Using Convolutional Neural Networks. *Iranian Conference on Signal Processing and Intelligent Systems*.

TEACHING

Lecturer: Artificial Intelligence in Robotics and Mechatronics, University of Tehran (Spring 2024–Spring 2025). Additional teaching: Python (2019–2020); teaching assistant in mechatronics, advanced robotics, mathematical engineering, and programming.

INVITED WORKSHOPS & PRESENTATIONS

- New Approaches to Robotic Grasping and Object Manipulation, 28th Elecomp Exhibition (2025).
- Human–Robot Interaction Laboratory: Research and Industrial Applications, AIoT Conference, Pardis Technology Park (2025).
- The Advancement of Robotic Manipulation Technologies: From Machine Learning to Industry 4.0, IranOpen (2025).
- New Approaches in Robotic Grasping, ICCKE (2024); Deep Learning Perspectives on Robotic Grasping, AISP (2024).

TECHNICAL SKILLS

Programming	Python, C++, MATLAB, R
Machine learning	PyTorch, TensorFlow, Keras, scikit-learn
Robotics	ROS, Gazebo, CoppeliaSim, PyBullet
Data systems	Apache Spark, Kafka, SQL, Oracle, Redis, MongoDB
Tools	Git, Jupyter, VS Code, LaTeX

HONORS & AWARDS

Ranked 14th, Alibaba ZhejiangLab Cup Global AI Competition, Zero-shot Learning Picture Recognition (2018; 3,225 teams). RoboCup Small Size League: 4th place (2017). IranOpen Small Size League: 1st technical challenge and 3rd overall (2017).

Full publication record: Google Scholar · Contact: hosseini.hamed@ut.ac.ir